

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12399476
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Johnson; Morgan et al.

General purpose command system and interface for a live broadcast

Abstract

A GPI device includes a communication port for receiving a GPI signal from a first device, one or more processors; and a memory comprising program instructions. The program instructions, when executed by the one or more processors, cause the GPI device to assign a command to the GPI signal when the GPI device receives the GPI signal from the first device and transmit the command to a second device. The transmission of the command may be triggered by the received GPI signal. The command may include a script, such as a python script. The command may also include triggering transmission of a live broadcast, switching between graphics during the live broadcast, or a combination thereof.

Inventors: Johnson; Morgan (Ponte Vedra Beach, FL), Walker; Barry (Ponte Vedra Beach, FL), Fegan; Jonathan (Ponte Vedra Beach, FL), Vo; Tung (Ponte Vedra Beach, FL)

Applicant: PGA TOUR Enterprises, LLC (Ponte Vedra Beach, FL)

Family ID: 1000008782904

Assignee: PGA TOUR Enterprises, LLC (Ponte Vedra Beach, FL)

Appl. No.: 18/949736

Filed: November 15, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20250155862 A1	May. 15, 2025

Related U.S. Application Data

us-provisional-application US 63599489 20231115

Publication Classification

Int. Cl.: G05B19/042 (20060101); G06F8/40 (20180101)

U.S. Cl.:

CPC G05B19/0423 (20130101); G06F8/40 (20130101);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6069663	12/1999	Bessel	348/441	H04N 21/4113
6345364	12/2001	Lee	713/324	G06F 1/3265
6438618	12/2001	Lortz	719/318	G06F 9/542
7024677	12/2005	Snyder	725/86	G11B 27/031
8539080	12/2012	Uluderya	709/225	G06F 9/5027
9067132	12/2014	Bergeron	N/A	G06F 3/016
11457049	12/2021	Kolbert-Hyle	N/A	H04L 65/611
11607601	12/2022	Lovell et al.	N/A	N/A
11745084	12/2022	Lovell et al.	N/A	N/A
11872464	12/2023	Vitti et al.	N/A	N/A
11986699	12/2023	Walker et al.	N/A	N/A
11998829	12/2023	Lovell et al.	N/A	N/A
12161913	12/2023	Walker et al.	N/A	N/A
12179085	12/2023	Vitti et al.	N/A	N/A
2001/0047394	12/2000	Kloba	709/217	G06F 16/9574
2002/0109710	12/2001	Holtz	N/A	G11B 27/034
2003/0023971	12/2002	Martinolich	725/32	H04N 21/235
2003/0046709	12/2002	Kuo-Wen	348/E7.071	H04N 21/4782
2004/0073912	12/2003	Meza	719/321	G06F 16/258
2005/0210419	12/2004	Kela	382/103	G06F 3/0484
2005/0278618	12/2004	Ogikubo	725/135	G11B 27/034
2005/0281204	12/2004	Karol	370/252	H04L 43/50
2008/0091856	12/2007	Blaisdell	710/63	H04L 41/0809
2010/0103324	12/2009	Maegawa	348/E5.057	H04N 5/268
2010/0115147	12/2009	Kim	710/14	G06F 13/10
2010/0142447	12/2009	Schlicht	370/328	H04W 4/20
2010/0285771	12/2009	Peabody	379/38	G08B 25/001
2011/0132987	12/2010	Hoeksel	235/492	H04L 63/0853
2013/0024584	12/2012	Marcus	710/63	G06F 21/00
2013/0054603	12/2012	Birdwell	707/738	G06F 16/2465
2013/0176216	12/2012	Ichieda	345/157	H04N 9/312
2014/0082117	12/2013	Unhale	709/208	H04L 67/306
2014/0259147	12/2013	L'Heureux	726/14	H04L 63/02

2014/0267195	12/2013	Enns	345/184	G06F 3/0334
2015/0003539	12/2014	Umezaki	375/240.28	H04N 19/42
2015/0085131	12/2014	Anderson	348/157	H04N 23/62
2015/0326446	12/2014	Koverman	709/224	H04L 41/0627
2015/0382208	12/2014	Elliott et al.	N/A	N/A
2016/0239819	12/2015	Yamada	N/A	G06Q 20/209
2016/0371099	12/2015	Woog	N/A	H04L 69/28
2017/0031689	12/2016	Wokhlu	N/A	G06F 9/3004
2017/0060541	12/2016	Saleh	N/A	G06F 8/34
2017/0115689	12/2016	Liu	N/A	G06F 1/1605
2017/0300369	12/2016	Cheung	N/A	G06F 16/27
2018/0013583	12/2017	Rubenstein	N/A	H04L 61/4511
2018/0034889	12/2017	Rubenstein	N/A	H04N 21/6125
2018/0091417	12/2017	Oré	N/A	H04L 45/302
2018/0189853	12/2017	Stewart	N/A	B01F 33/848
2018/0232323	12/2017	Ghosh	N/A	G06F 13/4291
2018/0254648	12/2017	Harju	N/A	H02J 7/00
2018/0270161	12/2017	Popescu	N/A	H04L 43/0864
2019/0129492	12/2018	Chen	N/A	G06F 1/266
2019/0244425	12/2018	Koohmarey	N/A	N/A
2020/0080741	12/2019	Lau	N/A	F24F 11/64
2020/0099735	12/2019	Mashaal	N/A	G06F 15/76
2020/0112167	12/2019	Jang	N/A	H02H 1/0007
2020/0112363	12/2019	Roy	N/A	H04W 40/12
2020/0117564	12/2019	Wu	N/A	G06F 11/2635
2020/0293549	12/2019	Wang	N/A	H04L 67/1095
2020/0409444	12/2019	Delshadpour	N/A	H04L 25/0298
2021/0216213	12/2020	Shveidel	N/A	H04L 67/1001
2021/0283498	12/2020	Zhang	N/A	G06F 9/547
2021/0339132	12/2020	Shakespeare	N/A	A63F 13/537
2021/0360295	12/2020	Sze	N/A	G11B 27/34
2021/0393122	12/2020	Milea	N/A	A61B 3/04
2021/0406079	12/2020	Atur	N/A	G06F 9/4881
2022/0165027	12/2021	Lee	N/A	G06V 10/803
2022/0189478	12/2021	Boo	N/A	G10L 17/22
2022/0393966	12/2021	Eriksson	N/A	H04L 45/24
2023/0238020	12/2022	Yae	N/A	G10L 25/84
2023/0334334	12/2022	Wang et al.	N/A	N/A
2024/0071220	12/2023	Perumalla	N/A	B60W 60/0015
2024/0316436	12/2023	Lovell et al.	N/A	N/A
2024/0325859	12/2023	Lovell et al.	N/A	N/A
2024/0362071	12/2023	Roque	N/A	G06F 9/5072
2024/0385767	12/2023	Desai	N/A	G06F 12/0246
2025/0099808	12/2024	Walker et al.	N/A	N/A

OTHER PUBLICATIONS

International Search Report issued Jan. 17, 2025 in connection with PCT Application No. PCT/US2024/056268. cited by applicant

Written Opinion of the International Searching Authority issued Jan. 17, 2025 in connection with PCT Application No. PCT/US2024/056268. cited by applicant

U.S. Appl. No. 18/987,874, filed Dec. 19, 2024 entitled Pan and Tilt Head. cited by applicant
U.S. Appl. No. 19/044,610, filed Feb. 3, 2025 entitled Event Data Immersion System. cited by applicant
U.S. Appl. No. 19/087,160, filed Mar. 21, 2025 entitled Golf Cart Tow Link. cited by applicant
U.S. Appl. No. 18/999,995, filed Dec. 23, 2024 entitled Golf Tournament Management System. cited by applicant
U.S. Appl. No. 18/963,118, filed Nov. 27, 2024 entitled Golf Play Outcome Simulation Modeling System. cited by applicant
U.S. Appl. No. 18/813,325, filed Aug. 23, 2024 entitled Golf Event Broadcast Production System. cited by applicant
U.S. Appl. No. 18/510,522, filed Nov. 15, 2023 entitled Hybrid Power Systems and Methods. cited by applicant
U.S. Appl. No. 18/949,970, filed Nov. 15, 2024 entitled Golf Tracking System for Monitoring and Management of Data. cited by applicant
U.S. Appl. No. 18/785,659, filed Jul. 26, 2024 entitled Platform to Manipulate Golf Data to Enable Creation of Broadcast Production Graphics. cited by applicant
U.S. Appl. No. 19/170,473, filed Apr. 4, 2025 entitled Management of Golf Tournament Scoring Data. cited by applicant

Primary Examiner: Lee; Adam

Attorney, Agent or Firm: Akerman LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application claims the benefit of the priority data of U.S. Provisional Patent Application No. 63/599,489, filed Nov. 15, 2023, the contents of which are hereby incorporated herein by reference.

TECHNOLOGY

(1) The present disclosure is directed to GPI command systems including a GPI device and methods of generating commands from GPI signals to control non-GPI compatible devices.

BACKGROUND

(2) A general purpose interface (GPI) is a signaling protocol used by a first device to control a second device. GPI generally speaking, GPI employs high/low voltage signaling and has historically enjoyed broad application. However, rapid technological advancements and diversification in both hardware and software have created complex interface compatibility issues that have rendered GPI signaling protocols unsuitable for use between many devices. What is needed are solutions to bridge the compatibility gap and increase the application of GPI to devices and software that are currently incompatible.

SUMMARY

(3) In one aspect, a GPI device includes a communication port for receiving a GPI signal from a first device, one or more processors; and a memory comprising program instructions. The program instructions, when executed by the one or more processors, cause the GPI device to assign a command to the GPI signal when the GPI device receives the GPI signal from the first device and transmit the command to a second device. The transmission of the command is triggered by the received GPI signal. In one example, the command includes a python script.

(4) In one example, the GPI device may include a single board computer compatible with the python script. The GPI device may be further configured to convert the GPI signal to the python

script.

- (5) In any combination of the above examples or another example, the command may include a signal to open or close a circuit or switch, power on, power off, trigger an actuator, modify power supply to control a motor, or a combination thereof.
- (6) In any combination of the above examples or another example, the command may include instructions for the second device to further modify the command.
- (7) In any combination of the above examples or another example, the command may include triggering transmission of a live broadcast, switching between graphics during the live broadcast, or a combination thereof. The GPI device may be further configured to utilize smart routing to transmit the GPI signal. Smart routing further comprises optimizing latency for instantaneous transmission and delivery of the command to the second device when the first device triggers transmission of the GPI signal that triggers the transmission of the command to the second device. The command or the script may further include routing data that specifies where the python script is to be transmitted when utilizing smart routing.
- (8) In one example, the first device comprises a broadcast switcher, the second device comprises a graphics engine. In a further example, the broadcast switcher and the command instruct the graphics engine to generate a graphic. In another example, the second device comprises a video server and the command instructs the video server to play a video.
- (9) In one aspect, a method of controlling multiple devices using a GPI device includes receiving, by a communication port, a GPI signal from a first device; assigning, using one or more processors, a command to the GPI signal; and transmitting, using the one or more processors, the command to a second device. Transmitting the command may be triggered by receiving the GPI signal. In one example, the command comprises python script.
- (10) In one example, the GPI device further includes a single board computer compatible with python script. In this aspect, method further includes converting the GPI signal to the python script.
- (11) In the above or another example, the command may include a signal to open or close a circuit or switch, power on, power off, trigger an actuator, modify power supply to control a motor, or a combination thereof. The command may also instruct the second device to further modify the command.
- (12) In any combination of the above examples or another example, the command includes triggering transmission of a live broadcast, switching between graphics during the live broadcast, or a combination thereof. The method may further include utilizing smart routing to transmit the command. In this aspect, smart routing further comprises optimizing latency for instantaneous transmission and delivery of the command to the second device when the first device triggers transmission of the GPI signal that triggers the transmission of the command to the second device. The command or the script further include routing data that specifies where the python script is to be transmitted.
- (13) In one example of the method, the first device is a broadcast switcher, the second device is a graphics engine and the method further includes instructing the graphics engine to generate a graphic. In another example of the method, the second device is a video server and the method further includes instructing the video server to play a video.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The novel features of the described embodiments are set forth with particularity in the appended claims. The described embodiments, however, both as to organization and manner of operation, may be best understood by reference to the following description, taken in conjunction with the accompanying drawings in which:

- (2) FIG. 1 schematically illustrates a GPI command system **10** including a GPI device according to various embodiments herein; and
- (3) FIG. 2 schematically illustrates a GPI command system **10** including a GPI device according to various embodiments herein.

DESCRIPTION

- (4) Various embodiments of a GPI command system including a GPI device and methods of generating commands from GPI signals are described herein. FIGS. 1-2 illustrate various embodiments of a GPI command system **10** including a GPI device **30**, wherein like numbers indicate like features.
- (5) The GPI device **30** may be configured to receive GPI signals **26** from a first device, which may also be referred to herein as a command device **20**, and use the GPI signals **26** to trigger a command **36** that is ultimately used to control another device, which may also be referred to herein as a controlled device **40**. The GPI device **30** may comprise an electronic device operable to receive GPI signals **26** and cause a command **36** assigned to the GPI signal **26** to be triggered. The GPI device **30** may be further configured to output the command **36** in a language compatible with the controlled device **40** to cause the controlled device **40** to execute all or a portion of the command **36**. In various embodiments, the GPI device **30** is or comprises a computer, micro controller, board-level controller, expansion card, or single board computer.
- (6) The GPI device **30** may comprise or operatively integrate with a computer comprising a processor **31** and memory **34** storing instructions that when executed by the processor **31** perform the operations of the GPI device **30**. In various embodiments, operations of the GPI device **30** may include communications within, between, or among networks or networked devices. For example, one or more of the command device **20**, GPI device **30**, or controlled device **40** may reside on one or more networks (e.g., WLAN, LAN, WAN, internet, or Cloud). The command device **20**, GPI device **30**, or controlled device **40** may be within the same or different networks. In an above or another example, the command device **20**, GPI device, controlled device or combination thereof may utilize a VPN to securely transmit respective GPI signals **26** or commands **36**. In some embodiments, one or more of the command device **20**, GPI device **30**, or controlled device **40** is not associated with a network.
- (7) In some embodiments, the memory **30** may be tangible, non-transitory computer-readable media including program instructions. The program instructions, when executed by processor **31**, cause the GPI device **30** to perform operations when the GPI device **30** receives the GPI signal **26** from the command device **20**. The operations may include assigning the command **36** to the GPI signal **26** and transmitting the command **36** to the controlled device **40**.
- (8) The GPI device **30** may include a communication port **32** configured to receive the GPI signals **26** from the command device **20** directly, indirectly, over one or more wired or wireless networks, or combination thereof via wired or wireless data protocols. The communication port **32** may also be configured to transmit the command **36** triggered by the GPI signals **26** directly, indirectly through one or more intermediate devices, over one or more wired or wireless networks, or combination thereof via wired or wireless data protocols. As described in more detail below, the GPI device **30** may include input pins **33**, e.g., general purpose input pins, configured to receive the GPI signals. In this way, a receiver portion of the communication port **32** may include input pins **33**. Input pins **33** may be physical pins, analogs defined in hardware or software, or both.
- (9) The command device **20** may be any electronic device configured to generate GPI level signals. The command device **20** may be an electronic device such as a computer, remote clicker or other transmitter, a broadcast switcher, or the like. The command device **20** may include a communication port comprising a transmitter configured for wired or wireless communication. For example, GPI signals **26** may be transmitted by the command device **20** via I/O, IP, TCP, TCP/IP, RF, short range RF (e.g., UWB, WiFi, Bluetooth protocol), or cellular.
- (10) The command device **20** may include a communication port **22** configured to transmit GPI

signals **26** via wired or wireless protocols. The GPI signals **26** may be transmitted directly to a suitable communication port **32** of the GPI device **30** or may be transmitted indirectly through one or more wired or wireless networks or intermediate devices. For example, the command device **20** may transmit GPI signals **26** along one or more wires connecting the command device **20** to the GPI device **30**. In one configuration, one or more wires are connected to one or more input pins **33** of the GPI device **30**. In another configuration, the GPI signals **26** are transmitted via wired or wireless protocol to the communication port **32** of the GPI device **30** and the GPI signals **26** are provided to a detection circuit. The GPI device **30** may be programmed to interpret the received GPI signals **26** as GPI input signals for purposes of triggering the programmed command **36** corresponding to the GPI signal **26**. In another example, the GPI device **30** includes a relay that is coupled to input pins **33** to relay the GPI signals **26** to the input pins **33**. In one embodiment, the command device **20** transmits GPI signals **26** wirelessly to a network from the communication port **22** that transmits the GPI signals **26** to the GPI device **30** via wired or wireless protocols. The GPI device **30** may be on the same or different network. In one configuration, the command device **20** may wirelessly transmit the GPI signal **26** directly or indirectly to a relay connected to input pins of the GPI device **30** to relay the GPI signal **26** at the pins. In one example, the GPI device **30** includes the relay and the wirelessly transmitted GPI signal **26** is received by the communication port via a wired or wireless connection. In one example, pins of the GPI device **30** may be configured is software for control remotely based on the GPI signals **26**.

(11) The controlled device **40** may be any electronic device capable of being controlled by a command **36**. For example, the command **36** may be received by the controlled device **40** as a signal to open or close a circuit or switch, power on, power off, trigger an actuator, modify power supply to control a motor, or any other commendable action. The command **36** may be executed in hardware or software. For example, the controlled device **40** may include a processor configured to execute the command **36**. The controlled device **40** may also include memory storing instructions that when executed by the processor cause the processor to perform the operations of the controlled device **40**, which may include executing the command **36**. For instance, the instructions may include a program wherein the command is in the same or a compatible language with the program such that the command **36** may be executed by the controlled device **40**. In one example, the controlled device **40** comprises a graphics engine and the command **36** instructs the graphics engine to generate a graphic. In another example, the controlled device **40** comprises a video server and the command **36** instructs the video server to play a video. The instruction or result of execution of the command **36** is not limited. For instance, the command **36** may instruct the controlled device **40** to further modify the command **36** or other data, transmit the command **36** or other data, take an action, or perform some other task in hardware or software. The command device **20** may include a communication port **42** configured for wired or wireless communication. For example, commands **36** may be transmitted by the controlled device **40** via I/O, IP, TCP, TCP/IP, RF, short range RF (e.g, UWB, WiFi, Bluetooth protocol), or cellular.

(12) In operation, a GPI signal **26** may be received at input pins **33**, e.g., general purpose input pins, or otherwise by the GPI device **30** such that the GPI device **30** detects the signal and causes the assigned command **36** to be triggered. In some embodiments, the GPI device **30** includes general purpose input/output (GPIO) pins. The state of one or more of the pins may be defined as input in hardware or software.

(13) As introduced above, the GPI device **30** may be operable to receive GPI level signals at input pins **33** or otherwise and, based on the GPI signals **26**, output commands **36** assigned to the GPI signal **26** in a language compatible with the controlled device **40**. For example, a GPI device **30**, such as a Raspberry Pi single board computer, compatible with python may include GPIO pins programmed to detect GPI level signals that specify trigger events having associated python scripts that are transmitted to the controlled device **40** compatible with python language commands **36** upon detection of the assigned GPI signal **26**. Thus, the GPI signals **26** may be used to trigger GPI

defined commands **36** in the form of python scripts.

(14) Further to the above, the GPI device **30** may be programmed to convert GPI level signals to python script. The GPI level signals may be used to trigger events that run a defined function. The function may include a python script. In one example, the GPI signal **26** may trigger a callback function programmed to execute an assigned command **36** comprising transmitting the python script wherein the command **36** and python script are specified by the GPI signal **26** value. In one embodiment, the function, e.g., the command **36** or python script, may include routing data that specifies where the script is to be transmitted. In another embodiment, all scripts are transmitted to the same hardware or software destination, e.g., address. Routing may be through wired or wireless communication over one or more networks (e.g., WLAN, LAN, WAN, same or different network as the GPI device **30**, command device, or both, Cloud) using suitable communication protocols (e.g., TCP/IP, POP, HTTP, SMTP, WIRE, OSI).

(15) As described above, the GPI device **30** may be configured to operate in a wide variety of GPI command system **10** variations.

(16) In FIG. **1**, the command device **20** and GPI device **30** are connected by one or more wires **23** suitable for conducting the GPI signal **26**, e.g., copper wire. The one or more wires **23** are coupled between the communication port **22** of the command device **20**, comprising a transmitter, and a receiver portion of the communication port **32** of the GPI device **30**, comprising input pins **33**, configured to receive the GPI signal **26**. The GPI device **30** is programmed to detect the GPI signal **26** and trigger a programmed command **36** determined from the GPI signal **26** received. The command **36** comprises a script in a language compatible with the controlled device **40**. The GPI device **30** is configured to transmit the command **36** via a wired or wireless protocol to the command device for execution of script. For example, the communication ports **32**, **42** may be in digital communication via one or more network cables. As another example, communication port **32** may transmit a RF signal including the command **36** that the controlled device **40** executes when received by communication port **42**.

(17) FIG. **2** illustrates a GPI command network **11** wherein the GPI signal **26** is used to trigger a command **36** to a cloud computing environment. The command device **20** and GPI device **30** may be connected by one or more wires **23** suitable for conducting the GPI signal **26**, e.g., copper wire, as described with respect to FIG. **1** or the GPI signal **26** may be transmitted to the GPI device **30** by wireless protocol. The GPI device **30** is programmed to detect the GPI signal **26** and trigger a programmed command **36** determined from the GPI signal **26** received. The command **36** comprises a script in a language compatible with the controlled device **40**. The GPI device **30** is in wired or wireless data communication with a network **52** that transmits the command **36** to the cloud **54** for execution of the command **36** to control the controlled device **40**. The connection with the network **52** may utilize TCP/IP connection protocol to pass the command **36**, e.g., message.

(18) In one embodiment, a method of triggering a command **36** from a command device **20** using GPI signals **26** to control a controlled device **40** includes receiving the GPI signals **26** at a GPI device **30** as described herein and using the GPI signals **26** to trigger the command **36** wherein the command **36** comprises a script programmed to be triggered by the GPI signals **26** and transmitting the command **36** to the control device **40**. The method may utilize hardware, software, or data communication protocols as described in FIGS. **1-2** or elsewhere herein.

(19) In one example, the command device **20** comprises a production switcher, such as a broadcast switcher, and the controlled device **40** comprises a graphics engine. The GPI device **30** may provide a physical interface for a cloud-based broadcasting network trigger. For example, the GPI device **30** may be utilized to trigger commands to a cloud broadcast service rather than a local set of broadcasting tools. The GPI device **30** may be employed as the last step in a hardware control chain between a video switcher command device **20** and the devices it communicates with via GPI before entering the cloud. The GPI device **30** may catch the broadcast style GPI signal **26** and convert it into a script, such as Python script, that can be transmitted anywhere, such as a laptop

sitting on the same LAN as the GPI device **30** or machine running on a cloud service, such as Amazon EC2, instance also accessible via the internet. In one such example, the GPI device **30** comprises a Raspberry Pi computer that connects to the video switcher command device **20** via low level electric contact closures on copper wires connected between the devices.

(20) In various embodiments, the GPI device is provided as a broadcast or graphics switching service for live broadcast, sports, or gambling. Accordingly, a method of broadcast or graphics switching for live broadcasts, sports, or gambling may include transmitting a GPI signal to the GPI device.

(21) In one embodiment, the GPI device, which may include a GPI service, may be used for triggering any mission critical function either in remote hardware or software that requires extreme latency. For example, the GPI device may include, be configured with, or be configured to transmit triggered commands via smart routing that optimizes latency for command data delivery for near instantaneous delivery of commands to a controlled device upon the command device triggering transmission of the GPI signal that triggers the transmission of the command to the controlled device. Smart routing may comprise core enhancements that ensure latency is sufficiently reduced to allow for use in a live broadcast, sports, gambling or other setting where trigger speed is imperative. In one example, smart routing may include employing continuously updated mapping of optimal data paths to network services to optimize latency for data delivery and triggering. In a further example, the updated mapping may be continually optimized in the cloud layer, at the trigger level, or in other locations or using different methods along the routing path. In a further example, this smart routing may be configured to utilize continuously provided network intelligence or previously defined optimal or reserved pathing. In various embodiments, the GPI device may be configured to utilize smart routing such that a user of a command device may push a button, flip a switch, or otherwise initiate transmission of a GPI signal and the triggered command nearly instantaneously is received by the controlled device to cause the desired action or result.

(22) In various embodiments, the GPI device may operate as a service that is administered through scripts stored in a cloud or other computing service that allows a service client computer to self-register. For example, a service client may register with the service and create or select desired scripting options. In a further example, the scripting options may be created or selected for use across multiple types of output trigger devices. Additionally or alternatively, a service client may register for use of existing scripting services. Thus, the service may be customizable by the service client such that GPI signals triggered by different command devices may be received to trigger desired commands, which may be the same command from a same GPI signal received from different command devices, different commands from the same GPI signal received from different command devices, the same commands from different GPI signals received from different command devices, different commands from different GPI signals received from the same command device, or different commands from different GPI signals received from different command devices. In one embodiment, the ability to push or send a command from one graphics device to another graphics device, which may be a physical graphics device or cloud-based, using the GPI device, or service thereof, is based on when a button or other defined start mechanism for the service is triggered for the relationship between the GPI service and the connected graphic devices.

(23) At least a portion of the systems, GPI device, methodologies, and techniques described with respect to the exemplary embodiments may incorporate a machine, such as, but not limited to a computer system or other computing device within which a set of instructions, when executed, may cause the machine to perform any one or more of the methodologies or functions discussed above. The machine may be configured to facilitate various operations conducted by the systems. For example, the machine may be configured to, but is not limited to, assist the systems by providing processing power to assist with processing loads experienced in the systems, by providing storage capacity for storing instructions or data traversing the systems, or by assisting with any other

operations conducted by or within the systems. In some embodiments, the machine may operate as a standalone device. In some embodiments, the machine may be connected (e.g., using communications network, another network, or a combination thereof) to and assist with operations performed by other machines and systems, such as, but not limited to, the command device, GPI device, controlled device, network, databases, any other system, program, and/or device, or any combination thereof. The machine may be connected with any component in the systems. In a networked deployment, the machine may operate in the capacity of a server or a client user machine in a server-client user network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine may comprise a server computer, a client user computer, a personal computer (PC), a tablet PC, a laptop computer, a desktop computer, a control system, a network router, switch or bridge, or any machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein.

(24) The computer system may include a processor (e.g., a central processing unit (CPU), a graphics processing unit (GPU), or both), a main memory and a static memory, which communicate with each other via a bus. The computer system may further include a video display unit, which may be, but is not limited to, a liquid crystal display (LCD), a flat panel, a solid state display, or a cathode ray tube (CRT). The computer system may include an input device, such as, but not limited to, a keyboard, a cursor control device, such as, but not limited to, a mouse, a disk drive unit, a signal generation device, such as, but not limited to, a speaker or remote control, and a network interface device. The disk drive unit may include a machine-readable medium on which is stored one or more sets of instructions, such as, but not limited to, software embodying any one or more of the methodologies or functions described herein, including those methods illustrated above. The instructions may also reside, completely or at least partially, within the main memory, the static memory, or within the processor, or a combination thereof, during execution thereof by the computer system. The main memory and the processor also may constitute machine-readable media.

(25) Dedicated hardware implementations including, but not limited to, application specific integrated circuits, programmable logic arrays and other hardware devices can likewise be constructed to implement the methods described herein. Applications that may include the apparatus and systems of various embodiments broadly include a variety of electronic and computer systems. Some embodiments implement functions in two or more specific interconnected hardware modules or devices with related control and data signals communicated between and through the modules, or as portions of an application-specific integrated circuit. Thus, the example system is applicable to software, firmware, and hardware implementations.

(26) In accordance with various embodiments of the present disclosure, certain methods described herein are intended for operation as software programs running on a computer processor. Furthermore, software implementations can include, but not limited to, distributed processing or component/object distributed processing, parallel processing, or virtual machine processing can also be constructed to implement the methods described herein.

(27) The present disclosure contemplates a machine-readable medium containing instructions so that a device connected to the communications network, another network, or a combination thereof, can send or receive voice, video or data, and communicate over the communications network, another network, or a combination thereof, using the instructions. The instructions may further be transmitted or received over the communications network, another network, or a combination thereof, via the network interface device. The term “machine-readable medium” should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) that store one or more sets of instructions. The term “machine-

readable medium” shall also be taken to include any medium that is capable of storing, encoding or carrying a set of instructions for execution by the machine and that causes the machine to perform any one or more of the methodologies of the present disclosure. The terms “machine-readable medium,” “machine-readable device,” or “computer-readable device” shall accordingly be taken to include, but not be limited to: memory devices, solid-state memories such as a memory card or other package that houses one or more read-only (non-volatile) memories, random access memories, or other re-writable (volatile) memories; magneto-optical or optical medium such as a disk or tape; or other self-contained information archive or set of archives is considered a distribution medium equivalent to a tangible storage medium. The “machine-readable medium,” “machine-readable device,” or “computer-readable device” may be non-transitory, and, in certain embodiments, may not include a wave or signal per se. Accordingly, the disclosure is considered to include any one or more of a machine-readable medium or a distribution medium, as listed herein and including art-recognized equivalents and successor media, in which the software implementations herein are stored.

(28) The illustrations of arrangements described herein are intended to provide a general understanding of the structure of various embodiments, and they are not intended to serve as a complete description of all the elements and features of apparatus and systems that might make use of the structures described herein. Other arrangements may be utilized and derived therefrom, such that structural and logical substitutions and changes may be made without departing from the scope of this disclosure. Figures are also merely representational and may not be drawn to scale. Certain proportions thereof may be exaggerated, while others may be minimized. Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense.

(29) Thus, although specific arrangements have been illustrated and described herein, it should be appreciated that any arrangement calculated to achieve the same purpose may be substituted for the specific arrangement shown. This disclosure is intended to cover any and all adaptations or variations of various embodiments and arrangements of the invention. Combinations of the above arrangements, and other arrangements not specifically described herein, will be apparent to those of skill in the art upon reviewing the above description. Therefore, it is intended that the disclosure not be limited to the particular arrangement(s) disclosed as the best mode contemplated for carrying out this invention, but that the invention will include all embodiments and arrangements falling within the scope of the appended claims.

(30) The foregoing is provided for purposes of illustrating, explaining, and describing embodiments of this invention. Modifications and adaptations to these embodiments will be apparent to those skilled in the art and may be made without departing from the scope or spirit of this invention.

Claims

1. A general purpose interface (GPI) device comprising: a communication port configured to receive a general purpose signal from a first device and transmit a command to a second device; a general purpose interface including general purpose input pins for detecting general purpose signals; one or more processors including a single board computer compatible with the command; and a memory comprising program instructions, wherein the program instructions, when executed by the one or more processors, configure the GPI device to: detect, at the general purpose interface, the general purpose signal transmitted by the first device; based on the detection: trigger the GPI device to convert the general purpose signal to the command assigned to the general purpose signal; and trigger a callback function to transmit the command to the second device via the communication port, wherein the command comprises a script in a language compatible with the second device, and wherein the second device executes the command.
2. The GPI device of claim 1, wherein the script is python script.
3. The GPI device of claim 1, wherein the command includes a signal for the second device to open

or close a circuit or switch, power on, power off, trigger an actuator, modify power supply to control a motor, or a combination thereof.

4. The GPI device of claim 1, wherein the command instructs the second device to further modify the command.
 5. The GPI device of claim 1, wherein the command triggers the second device to transmit a live broadcast, switch between graphics during the live broadcast, or a combination thereof.
 6. The GPI device of claim 5, wherein the GPI device is further configured to utilize smart routing to transmit the general purpose signal and/or the command.
 7. The GPI device of claim 6, wherein the callback function further includes routing data that specifies where the script is to be transmitted.
 8. The GPI device of claim 6, wherein the smart routing comprises optimizing latency for near instantaneous transmission and delivery of the command to the second device when the detection triggers the callback function to transmit the command to the second device.
 9. The GPI device of claim 1, wherein the first device is a broadcast switcher, wherein the second device is a graphics engine, and wherein the command instructs the graphics engine to generate a graphic.
 10. The GPI device of claim 1, wherein the second device is a video server, and wherein the command instructs the video server to play a video.
 11. A method comprising: receiving, at a communication port of a general purpose interface (GPI) device, a general purpose signal from a first device; detecting, at a general purpose interface including general purpose input pins of the GPI device, the general purpose signal transmitted to the GPI device by the first device; based on the detecting: triggering the GPI device to convert, using one or more processors of the GPI device, the general purpose signal to a command assigned to the general purpose signal; and triggering a callback function to transmit, using the communication port, the command to the second device, wherein: the one or more processors include a single board computer compatible with the command, the command comprises a script in a language compatible with the second device, and the second device executes the command.
 12. The method of claim 11, wherein the script comprises a python script.
 13. The method of claim 11, wherein the command includes a signal for the second device to open or close a circuit or switch, power on, power off, trigger an actuator, modify power supply to control a motor, or a combination thereof.
 14. The method of claim 11, wherein the command instructs the second device to further modify the command.
 15. The method of claim 11, wherein the command triggers the second device to transmit a live broadcast, switch between graphics during the live broadcast, or a combination thereof.
 16. The method of claim 15, wherein the method further comprises the GPI device utilizing smart routing to transmit the general purpose signal and/or the command.
 17. The method of claim 16, wherein the callback function includes routing data that specifies where the script is to be transmitted.
 18. The method of claim 16, wherein the smart routing further comprises optimizing latency for near instantaneous transmission and delivery of the command to the second device when the detecting triggers the callback function to transmit the command to the second device.
 19. The method of claim 11, wherein the first device is a broadcast switcher, wherein the second device is a graphics engine, and wherein the command instructs the graphics engine to generate a graphic.
 20. The method of claim 11, wherein the second device is a video server, and wherein the command instructs the video server to play a video.
-